# jQuery Introduction (Detailed Guide for PDF)

## What is jQuery?

jQuery is a fast, lightweight JavaScript library designed to simplify HTML DOM manipulation, event handling, animations, and AJAX.

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## History of jQuery

- Created by John Resig in 2006
- Designed to simplify JavaScript coding
- Still used in many legacy and simple projects

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## Why Use jQuery?

- Write less code ("Write less, do more")
- Cross-browser compatibility
- Easy DOM manipulation
- Simplifies AJAX and animations

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## Features of jQuery

- DOM manipulation
- Event handling
- Effects and animations
- AJAX support
- Cross-browser support

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## Adding jQuery

### Using CDN

```
<script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
```

Local File

```
<script src="jquery.js"></script>
```

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# jQuery Syntax

```
$(selector).action();
```

- $ → jQuery function
- selector → selects HTML element
- action → operation to perform

Example:

```
$("p").hide();
```

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# jQuery Selectors (Detailed)

- ID Selector → $("#id")
- Class Selector → $(".class")
- Element Selector → $("p")
- Multiple Selector → $("h1, p")

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# Document Ready Function

Ensures code runs after page loads:

```
$(document).ready(function() {
  // code here
});
```

Short form:

```
$(function() {
  // code
});
```

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# Event Handling (Detailed)

Common events:

- click
- dblclick
- mouseenter
- mouseleave
- keypress

Example:

```
$("button").click(function() {
  alert("Clicked");
});
```

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# Effects (Detailed)

```
$("p").hide();
$("p").show();
$("p").toggle();
$("p").fadeIn();
$("p").fadeOut();
$("p").slideUp();
$("p").slideDown();
```

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# Animations

```
$("div").animate({
  left: '250px',
  opacity: 0.5
});
```

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# DOM Manipulation (Advanced)

```
$("#demo").text("Hello");
$("#demo").html("<b>Hello</b>");
$("#demo").val("Input value");
```

Adding elements:

```
$("p").append("New Text");
$("p").prepend("Start Text");
```

Removing elements:

```
$("p").remove();
```

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# CSS Manipulation

```
$("p").css("color", "red");
$("p").addClass("myClass");
$("p").removeClass("myClass");
```

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# AJAX (Detailed)

```
$.ajax({
  url: "data.txt",
  method: "GET",
  success: function(result) {
    $("#demo").html(result);
  },
  error: function() {
    alert("Error loading data");
  }
});
```

Shortcut:

```
$("#demo").load("data.txt");
```

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# Chaining

Multiple actions in one line:

```
$("p").css("color", "red").slideUp(2000).slideDown(2000);
```

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# Example Project

```
<button id="btn">Click Me</button>
<p id="demo"></p>

<script>
$(document).ready(function() {
  $("#btn").click(function() {
    $("#demo").text("Hello jQuery!");
  });
});
</script>
```

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# Advantages of jQuery

- Easy to use
- Saves time
- Cross-browser support
- Large plugin library

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## Disadvantages of jQuery

- Less popular in modern frameworks
- Slower than pure JavaScript in some cases

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## Best Practices

- Use latest version
- Minimize usage in modern apps
- Combine with JavaScript
- Use CDN for faster loading

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